

For example, if the array $ar = [1, 2, 3]$, $1 + 2 + 3 = 6$, so return 6.

Function Description

Complete the `simpleArraySum` function with the following parameter(s):

- $ar[n]$: an array of integers

Returns

- int : the sum of the array elements

Input Format

The first line contains an integer, n , denoting the size of the array.
The second line contains n space-separated integers representing the array's elements.

Constraints

$$0 < n, ar[i] \leq 1000$$

Sample Input

STDIN	Function
6	<code>ar[]</code> size $n = 6$
1 2 3 4 10 11	<code>ar = [1, 2, 3, 4, 10, 11]</code>

Sample Output

31

Explanation

Print the sum of the array's elements: $1 + 2 + 3 + 4 + 10 + 11 = 31$.

Line: 20 Col: 1

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You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Input (stdin)

Download

16

21 2 3 4 10 11

Your Output (stdout)

Download

131

Expected Output

Download

131

An array is a data structure that stores elements of the same type in a contiguous block of memory. In an array, A , of size N , each memory location has some unique index, i (where $0 \leq i < N$), that can be referenced as $A[i]$ or A_i .

Your task is to reverse an array of integers.

Note: If you've already solved our C++ domain's Arrays Introduction challenge, you may want to skip this.

Example

$A = [1, 2, 3]$

Return $[3, 2, 1]$.

Function Description

Complete the function `reverseArray` with the following parameter(s):

- `int A[n]`: the array to reverse

Returns

- `int[n]`: the reversed array

Input Format

The first line contains an integer, N , the number of integers in A .

The second line contains N space-separated integers that make up A .

Constraints

- $1 \leq N \leq 10^3$
- $1 \leq A[i] \leq 10^4$, where $A[i]$ is the i^{th} integer in A

Sample Input 1

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```
4
1 2 3 4
```

4

Line: 25 Col: 1

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You have earned 10.00 points!

You are now 20 points away from the 1st star for your problem solving badge.

33%

10/30

Congratulations

You solved this challenge. Would you like to challenge your friends?



Next Challenge

Test case 0

Test case 1

Test case 2

Test case 3

Compiler Message

Success

Hidden Test Case

Unlock this testcase for 5 hackos.

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Explanation 0

In this example:

- $a = (a[0], a[1], a[2]) = (5, 6, 7)$
- $b = (b[0], b[1], b[2]) = (3, 6, 10)$

Now, let's compare each individual score:

- $a[0] > b[0]$, so Alice receives 1 point.
- $a[1] = b[1]$, so nobody receives a point.
- $a[2] < b[2]$, so Bob receives 1 point.

Alice's comparison score is 1, and Bob's comparison score is 1. Thus, we return the array `[1, 1]`.

Sample Input 1

17 28 30
99 16 8

Sample Output 1

2 1

Explanation 1

Comparing the 0th elements, `17 < 99` so Bob receives a point.

Comparing the 1st and 2nd elements, `28 > 16` and `30 > 8` so Alice receives two points.

The return array is `[2, 1]`.

32

Line: 32 Col: 1

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Sample Test case 0

Input (stdin)

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Sample Test case 1

1 5 6 7
2 3 6 10

Your Output (stdout)

1 1 1

Expected Output

Download

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Sample Input

```
1 1 1 0 0 0
0 1 0 0 0 0
1 1 1 0 0 0
0 0 2 4 4 0
0 0 0 2 0 0
0 0 1 2 4 0
```

Sample Output

19

Explanation

`arr` contains the following hourglasses:

```
1 1 1 1 1 0 1 0 0 0 0
1
1 1 1 1 1 0 1 0 0 0 0
0 1 0 1 0 0 0 0 0 0 0
1
0 0 2 0 2 4 2 4 4 4 0
1 1 1 1 1 0 1 0 0 0 0
0 0 2 4 4
0 0 0 0 2 0 0 2 0 0
0 0 2 0 2 4 2 4 4 4 0
0
0 0 1 0 1 2 1 2 4 2 4 0
```

The hourglass with the maximum sum (19) is:

```
2 4 4
2
1 2 4
```

34

Line: 34 Col: 1

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Sample Test case 0

Input (stdin)

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Sample Test case 1

Sample Test case 2

```
1 1 1 1 0 0 0
0 1 0 0 0 0
1 1 1 0 0 0
0 0 2 4 4 0
0 0 0 2 0 0
0 0 1 2 4 0
```

Your Output (stdout)

```
19
```

Expected Output

```
19
```

Download

Initial Values:
 $n = 2$
 $lastAnswer = 0$
 $arr[0] = []$
 $arr[1] = []$

Query 0: Append 5 to $arr[(0 \oplus 0) \% 2] = arr[0]$.
 $lastAnswer = 0$
 $arr[0] = [5]$
 $arr[1] = []$

Query 1: Append 7 to $arr[(1 \oplus 0) \% 2] = arr[1]$.
 $arr[0] = [5]$
 $arr[1] = [7]$

Query 2: Append 3 to $arr[(0 \oplus 0) \% 2] = arr[0]$.
 $lastAnswer = 0$
 $arr[0] = [5, 3]$
 $arr[1] = [7]$

Query 3: Assign the value at index 0 of $arr[(1 \oplus 0) \% 2] = arr[1]$ to $lastAnswer$. Store $lastAnswer$ in your answer array. $lastAnswer = 7$
 $arr[0] = [5, 3]$
 $arr[1] = [7]$

Query 4: Assign the value at index 1 of $arr[(1 \oplus 7) \% 2] = arr[0]$ to $lastAnswer$. Store $lastAnswer$ in your answer array. $lastAnswer = 3$
 $arr[0] = [5, 3]$
 $arr[1] = [7]$

Return your answer array [7, 3]. The code stub prints its elements on separate lines.

31

Line: 31 Col: 1

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Sample Test case 0

Input (stdin)

Download

```
1 2 5
2 1 0 5
3 1 1 7
4 1 0 3
5 2 1 0
6 2 1 1
```

Your Output (stdout)

```
1 7
2 3
```

Expected Output

Download

Return

- *long*: the sum of the array elements

Input Format

The first line of the input consists of an integer n .

The next line contains n space-separated integers contained in the array.

Output Format

Return the integer sum of the elements in the array.

Constraints

$1 \leq n \leq 10$

$0 \leq arr[i] \leq 10^{10}$

Sample Input

STDIN	Function
5	arr[] size n = 5
1000000001 1000000002 1000000003 1000000004 1000000005	arr[]

Output

5000000015

Note:

The range of the 32-bit integer is (-2^{31}) to $(2^{31} - 1)$ or $[-2147483648, 2147483647]$.

When we add several integer values, the resulting sum might exceed the above range. You might need to use long int C/C++/Java to store such sums.

Line: 20 Col: 1

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✔ Sample Test case 0

Input (stdin)

Download

1 5

2 1000000001 1000000002 1000000003 1000000004 1000000005

Your Output (stdout)

1 5000000015

Expected Output

1 5000000015

Download

Returns

- `int[n]`: the rotated array

Input Format

The first line contains two space-separated integers that denote n , the number of integers, and d , the number of left rotations to perform.

The second line contains n space-separated integers that describe `arr[]`.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq d \leq n$
- $1 \leq a[i] \leq 10^6$

Sample Input

STDIN	Function
5 4	<code>n = 5 d = 4</code>
1 2 3 4 5	<code>arr = [1, 2, 3, 4, 5]</code>

Sample Output

5 1 2 3 4

Explanation

To perform $d = 4$ left rotations, the array undergoes the following sequence of changes:

`[1, 2, 3, 4, 5] → [2, 3, 4, 5, 1] → [3, 4, 5, 1, 2] → [4, 5, 1, 2, 3] → [5, 1, 2, 3, 4]`

26

Line: 26 Col: 1

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✔ Sample Test case 0

Input (stdin)

1 5 4

2 1 2 3 4 5

Your Output (stdout)

1 5 1 2 3 4

Expected Output

1 5 1 2 3 4

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STDIN	Function
3	arr[][] sizes n = 3, m = 3
11 2 4	arr = [[11, 2, 4], [4, 5, 6], [10, 8, -12]]
4 5 6	
10 8 -12	

Sample Output

15

Explanation

The primary diagonal is:

11
5
-12

Sum across the primary diagonal: $11 + 5 - 12 = 4$.

The secondary diagonal is:

4
5
10

Sum across the secondary diagonal: $4 + 5 + 10 = 19$

Difference: $|4 - 19| = 15$

Note: $|x|$ is the [absolute value](#) of x .

```
22         secondary += arr[i][j];
23     }
24 }
25 }
26
27     cout << abs(primary - secondary) << endl;
```

Line: 31 Col: 1

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Run Code

Submit Code

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Sample Test case 0

Input (stdin)

1 3
2 11 2 4
3 4 5 6
4 10 8 -12

Download

Your Output (stdout)

1 15

Expected Output

Download

Sample Input 3

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```
13
abcde
sdaklfj
asdjf
na
basdn
sdaklfj
asdjf
na
asdjf
na
basdn
sdaklfj
asdjf
5
abcde
sdaklfj
asdjf
na
basdn
```

```
abcde sdaklfj asdjf na basdn
Array: queries
```

Sample Output 3

```
1
3
4
3
2
```

30

Line: 30 Col: 1

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Run Code

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Sample Test case 0

Input (stdin)

Download

Sample Test case 1

Sample Test case 2

```
1 4
2 aba
3 baba
4 aba
5 xzxb
6 3
7 aba
8 xzxb
9 ab
```

Your Output (stdout)

```
1 2
```

Print the ratios of positive, negative and zero values in the array. Each value should be printed on a separate line with 6 digits after the decimal. The function should not return a value.

Input Format

The first line contains an integer, *n*, the size of the array.
The second line contains *n* space-separated integers that describe *arr[n]*.

Constraints

$0 < n \leq 100$
 $-100 \leq arr[i] \leq 100$

Sample Input

STDIN	Function
6	arr[] size n = 6
-4 3 -9 0 4 1	arr = [-4, 3, -9, 0, 4, 1]

Sample Output

0.500000
0.333333
0.166667

Explanation

There are 3 positive numbers, 2 negative numbers, and 1 zero in the array.
The proportions of occurrence are positive: $\frac{3}{6} = 0.500000$, negative: $\frac{2}{6} = 0.333333$
and zeros: $\frac{1}{6} = 0.166667$.

28

Line: 28 Col: 1

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Sample Test case 0

Input (stdin)

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Sample Test case 1

1 6
2 -4 3 -9 0 4 1

Your Output (stdout)

1 0.500000
2 0.333333
3 0.166667

Expected Output

1 0.500000
2 0.333333
3 0.166667

Download

the machine contains two space-separated integers n and q , the size of the array and the number of queries.

Each of the next q lines contains three space-separated integers a , b and k , the left index, right index and number to add.

Constraints

- $3 \leq n \leq 10^7$
- $1 \leq m \leq 2 * 10^5$
- $1 \leq a \leq b \leq n$
- $0 \leq k \leq 10^9$

Sample Input

STDIN	Function
5 3	arr[] size n = 5, queries[] size q = 3
1 2 100	queries = [[1, 2, 100], [2, 5, 100], [3, 4, 100]]
2 5 100	
3 4 100	

Sample Output

200

Explanation

After the first update the list is 100 100 0 0 0.

After the second update list is 100 200 100 100 100.

After the third update list is 100 200 200 200 100.

The maximum value is 200.

36

Line: 35 Col: 1

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Sample Test case 0

Input (stdin)

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Sample Test case 1

1 5 3
2 1 2 100
3 2 5 100
4 3 4 100

Sample Test case 2

Your Output (stdout)

1 200

Expected Output

1 200

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```
1 2
2 1
```

Sample Output 0

```
2
```

Explanation 0

The distance between points (1, 2) and (2, 1) is $\rho(1, 2) + \rho(2, 1) = 2$.

Sample Input 1

```
7 3
1 2
2 3
3 4
4 5
5 6
6 7
3 6
4 5
5 5
```

Sample Output 1

```
3
```

Explanation 1

The best points are (3, 6) and (5, 5), which gives us a distance of $\rho(3, 5) + \rho(6, 5) = 2 + 1 = 3$.

97

Line: 97 Col: 1

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Sample Test case 0

Input (stdin)

Download

Sample Test case 1

```
1 2 2
2 1 2
3 1 2
4 2 1
```

Your Output (stdout)

```
1 2
```

Expected Output

```
1 2
```

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